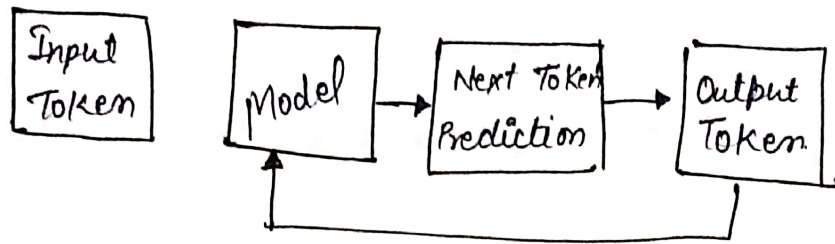


* Why ChatGPT Gets faster While Responding.

Basic Flow



@coddy d'di

② Why it feels slow at the start? warmup phase

- This step is heavy → More Computation
- It needs to process all input tokens to build context.

③ The Secret : KV Cache (key : Value Cache)

Transformers use Attention Mechanism.

For every token, model creates:

→ key (K) → Value (V)

Normally, for every new token, model would recompute K and V for all previous token

But with **KV Cache** : → it stores K and V reuses them.

KV Cache saves huge amount of recomputation → that's why it gets faster.

* Only New tokens is processed!

* Work stays almost constant. $O(n)$

④ Better Next Token Prediction

* As more tokens are generated, model get more context.

@codipolici

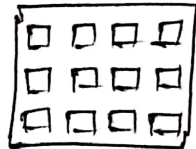
* More Context = clearer understanding.

* clearer understanding = higher confidence in next token.

* Higher confidence = faster Generation.

⑤ Parallel Processing on GPUs.

GPU are built for parallelism.



→ Many matrix operation happens in parallel.

→ After Warm Ups, GPU keeps the pipeline full.

⑥ Streaming makes it feel Even faster!

→ AI doesn't wait for full answer.

→ As soon as each token is ready, it is sent to you.

→ As you see the answer in real time.

⑦

Warm-Up

KV Cache

Better Predict

GPU parallel

Streaming output